

Inspect Before Modeling

Project: Responsible Plant Classifier

1. Inspect Before Modeling

Teach the learning cycle

Primary target: Inspect row meaning missing values and repeated records

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-4/5, ai-4/7, ai-4/9. Reuse a familiar case for recall

or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Python and Table Readiness

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

Look for: Specify a table schema

Reference example, not the only acceptable wording: activity: nonempty text; minutes: integer from 0 to 60 inclusive.

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

Look for: Separate prediction inputs from observed targets

Reference example, not the only acceptable wording: Candidate features are pages and queue_at_checkout. The target is returned_late and the tracking field is loan_id. days_to_return is known only after return and directly defines the target, so it must not enter checkout predictions.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

Concept to explain

Dataset shape, types, ranges, duplicates, and missingness determine whether modeling results can be trusted.

Term · Meaning

Term: shape

Meaning: number of rows and columns

Term: range

Meaning: smallest to largest plausible value

Term: duplicate

Meaning: repeated example or record

Term: missingness

Meaning: where values are absent

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach inspection before modelling in two 45-minute sessions. Use fictional data only. Emphasize the unit of observation, provenance, units, unknown values and overlapping audit flags. The dataset card is a concise classroom documentation exercise inspired by datasheets for datasets, not a claim to implement every part of a research framework. Students must distinguish observations from unsupported repairs and from evidence of model fitness.

Readiness answer

It is a numerical quantity whose differences measure time. Whole-number formatting does not turn it into a category.

Printed lesson and workbook

Cards for the five raw rows including both B rows

Optional Python 3 editor; no packages required

Blank card headings: purpose source row schema findings changes limits version owner

Apply the connected idea: Units conversions and data provenance

Teach Week 15 as a two-session investigation using fictional data. Two fictional activity logs use different duration units. Log A records 180 seconds and Log B records 4 minutes. Keep the raw values, units and source labels; derive a shared unit so another learner can reproduce the comparison. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

3 minutes, since 60 seconds make one minute.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can explain inspect before modeling using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Produce a data-quality report before fitting any model.

MODEL IDEA Dataset shape, types, ranges, duplicates, and missingness determine whether modeling results can be trusted.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Dataset shape is 40 rows × 5 columns. Inspection finds 3 duplicate rows and 4 missing light values.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Dataset shape is 40 rows × 5 columns.

STEP 2 Inspection finds 3 duplicate rows and 4 missing light values.

STEP 3 Leaf widths range from 0.4 to 48 cm; inspect the extreme value and unit.

STEP 4 Document, repair, or exclude by a stated rule—never silently delete inconvenient rows.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Lengths include 5 cm, blank and -3 cm, plus a duplicate photo row.

Name checks and a deletion risk.

Exact explanation to model

Check missingness, physical validity, units and duplicate identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Explain the added worked example

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before fitting a model, inspect what one row represents and document where the table came from. A dataset card is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes. Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no evidence about outdoor performance.

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition. We need evidence before deciding what happened.

A set stores each distinct ID once. The output is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs. This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full schema validation. It does not check text values, units, source truth or whether IDs refer to real unique events. A successful run does not certify data quality.

Key terms

Term · Meaning

Term: Provenance

Meaning: Where data came from and how it was created

Term: Schema

Meaning: The stated meaning types units and rules for fields

Term: Missing value

Meaning: An unknown or unavailable value rather than a measured zero

Term: Dataset card

Meaning: A concise record of dataset contents creation intended use and limitations

Apply the connected idea: Units conversions and data provenance

Two fictional activity logs use different duration units. Log A records 180 seconds and Log B records 4 minutes. Keep the raw values, units and source labels; derive a shared unit so another learner can reproduce the comparison.

The script prints [3.0, 4]. The slash operator returns a floating-point result. The original dictionaries are unchanged. Assume nonnegative numeric raw values have already been validated. An unsupported unit produces None here and must be reported for review rather than included in arithmetic.

1.5 minutes and 0.0 minutes. Two minutes is longer. Exit: No. Mark it unresolved and ask for the unit definition. Do not include it as a known duration.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip. Record the five-row count, the ID field, minutes as the duration unit and None as unknown. List the duplicate-ID and invalid-duration findings. State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add intended use, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice. Do not describe it as validated navigation data. For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Inspect Before Modeling

Spot a duplicate, an impossible value and a valid boundary before fitting.

The facts we will use

Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Inspect Before Modeling

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count rows

Five raw rows represent four distinct IDs; B is duplicated.



2. Check schema

C violates the nonnegative-height rule; D=0 is valid under this contract.



3. Flag before fitting

Flag duplicate B and invalid C. Do not count B twice or silently replace C.

CHANGE ONE THING

The teacher confirms B's second row is an accidental duplicate.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Count rows

What happens and why: Five raw rows represent four distinct IDs; B is duplicated.

Step: 2. Check schema

What happens and why: C violates the nonnegative-height rule; D=0 is valid under this contract.

Step: 3. Flag before fitting

What happens and why: Flag duplicate B and invalid C. Do not count B twice or silently replace C.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Flag duplicate B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A plausible nonnegative value can still be incorrectly measured; schema checks do not establish truth.

Change one thing in the pictured case

Change: The teacher confirms B's second row is an accidental duplicate.

Prediction: Remove that duplicate with a recorded reason; C still needs correction or explicit exclusion.

Reason: Resolving one defect does not resolve a separate impossible measurement.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Lengths include 5 cm, blank and -3 cm, plus a duplicate photo row.

Write the question to explain: Name checks and a deletion risk.

Reveal after predictions: Check missingness, physical validity, units and duplicate identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Keep for an independent attempt: A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: 2 of 5 raw rows have unknown duration, or 40 percent. No: the extra B occurrence is already among the missing rows, so subtracting all three flag counts double-counts it. A distinct usable-trip count requires a stated resolution policy and source review. Exit examples include duplicate IDs, impossible values, units, provenance and coverage. A defensible statement is that this five-row fictional export has documented quality flags and is suitable for inspection practice, with no model-readiness claim.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve regression target meaning and why later outcomes are separate from prediction inputs. Establish that this week inspects data before fitting.

Stage: 7 to 18 minutes

Teacher action: Read the source description and schema. Count physical rows and distinct trip IDs separately. Ask whether a repeated ID proves the reason for duplication.

Stage: 18 to 32 minutes

Teacher action: Trace the code with row cards. Explain set, None and negative checks. Complete W1 and W2 with named denominators.

Stage: 32 to 45 minutes

Teacher action: Use the guided question to show overlap: place both B cards in missing and repeated-ID groups. Do not subtract all flag counts independently.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve raw version versus corrected version. Ask what evidence could justify changing a recorded value.

Stage: 8 to 20 minutes

Teacher action: Read the card example and complete W3. Distinguish a documented fact, an unknown and an assumption.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently. Require counts, overlap and a concise card with source and limitations before offering hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit claim. Use the fresh retry after support and record the support given.

Teach the connected idea

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion task: Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

Teacher check: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The card records evidence and gaps; its claims still need checking. Documentation does not prove suitability.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Units conversions and data provenance

Target: Convert compatible units before comparing

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Convert 150 seconds and 30 seconds to minutes without rounding to whole numbers.

Teacher check: 2.5 and 0.5 minutes.

Individual check: For Units conversions and data provenance, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: Correct conversion does not establish that the sources measured the same activity, used the same start and stop rules, or are accurate. Provenance helps someone investigate those differences.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Keep raw row count 5, distinct IDs 4, duplicate B and invalid C, plus any confirmed correction and its reason. Changed-case check: Remove that duplicate with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

Changed-case reasoning: Remove that duplicate with a recorded reason; C still needs correction or explicit exclusion. Resolving one defect does not resolve a separate impossible measurement.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Data Triage: calculate missing counts, flag ranges, find duplicates, and write a keep/repair/exclude reason for each issue.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A model will automatically ignore harmless data-quality problems.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A complete card certifies data quality: The card records evidence and gaps; its claims still need checking. Documentation does not prove suitability.

Missing means zero: None is unknown; zero is a specific allowed value in this exercise. Substitution without evidence changes the meaning.

Flag counts can always be added: The same row can have multiple flags. Count rows or unique events under an explicit policy and retain the raw denominator.

Check these misconceptions

A successful run proves the conclusion: Correct conversion does not establish that the sources measured the same activity, used the same start and stop rules, or are accurate. Provenance helps someone investigate those differences.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: 90 seconds, 2 minutes and 30 seconds become 1.5, 2, 0.5; total 4 minutes. Keep the three source records.

A convenient shortcut preserves meaning: Comparing raw 180 to raw 4 ignores their units. Dividing 4 by 60 without checking its unit is also wrong: it was already minutes. Branch on the documented unit and keep a conversion note.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Check missingness, physical validity, units and duplicate identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Independent question

A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.

Expected independent answer

The negative height is implausible for this measurement and the blank is missing. Check source records and units before choosing a documented correction or exclusion.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Replace every suspicious value with zero without investigating.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Check -2 against source and units because physical length is nonnegative; mark missing explicitly; identify the duplicate and preserve provenance; keep 8 if valid. Document repairs and avoid automatically deleting conditions disproportionately represented among problematic rows.

Explanations for the additional workbook examples

W1 Define the row and source

For Robot practice v1, state what one row is intended to represent, the source, the unit and the meaning of None.

One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

W2 Interpret the audit counts

Explain each number in 5 4 2 1. How many extra ID occurrences are there, and how many rows belong to a repeated-ID group?

The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

W3 Reject an unsupported correction

A learner changes None to 0 and -2 to 2, then reports a clean dataset. Explain why this is not a justified repair and what record should be kept.

Those edits invent outcomes without source evidence. Preserve the raw version, flag the affected rows and investigate. Record any supported correction with old value, new value, reason and version; unresolved values remain uncertain.

W4 Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same schema. Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short dataset card stating source, row meaning, unit, findings, unchanged version and one excluded use.

There are 6 rows, 5 distinct IDs, 2 missing durations, 1 negative duration and 1 extra ID occurrence. Both Q rows are missing, so the duplicate and missing flags overlap. Zero for T is allowed by the stated schema. The card should identify the supplied fictional indoor export, one intended row per trip, minutes, all findings, raw unchanged version and no evidence for real or outdoor navigation use.

W5 State an evidence gap

Someone duplicates each row ten times and claims the export now covers more kinds of trips. Explain what changed and what did not.

The row count increased but no new trip conditions or independent observations were added. Coverage remains limited to the same supplied fictional indoor cases; the larger count does not establish broader usefulness.

Units conversions and data provenance — W1 Retain the source

Name four pieces of information to keep for Log A and its conversion.

Source A, raw 180, unit seconds and derived 3.0 minutes; also record divide by 60.

Units conversions and data provenance — W2 Convert precisely

Convert 150 seconds and 30 seconds to minutes without rounding to whole numbers.

2.5 and 0.5 minutes.

There are 60 seconds in one minute. Divide each seconds value by 60: $150/60 = 2.5$ minutes and $30/60 = 0.5$ minutes. Do not round away the half-minute. Check the direction by converting back: 2.5 times 60 is 150.

Units conversions and data provenance — W3 Repair a comparison

A report claims 120 seconds exceeds 3 minutes because $120 > 3$. Correct it.

120 seconds is 2 minutes, so 3 minutes is longer.

First put both values in the same unit. 120 seconds divided by 60 is 2 minutes. Now compare 2 minutes with 3 minutes: 3 minutes is longer by 1 minute. Comparing the raw numbers 120 and 3 without their units compares unlike quantities.

Units conversions and data provenance — W4 Normalize independently

Create records for 240 seconds, 1.5 minutes and 60 seconds. Derive minutes while keeping raw data. State the values and their total.

4.0, 1.5 and 1.0 minutes; total 6.5 minutes. Retain original units and values.

Units conversions and data provenance — W5 Handle an unknown unit

A record contains value 8 and unit ticks with no conversion rule. Should you assume seconds?

No. Mark it unresolved and ask for the unit definition. Do not include it as a known duration.

Assess independent transfer

Selected case: A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same schema. Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short dataset card stating source, row meaning, unit, findings, unchanged version and one excluded use.

Selected case answer

There are 6 rows, 5 distinct IDs, 2 missing durations, 1 negative duration and 1 extra ID occurrence. Both Q rows are missing, so the duplicate and missing flags overlap. Zero for T is allowed by the stated schema. The card should identify the supplied fictional indoor export, one intended row per trip, minutes, all findings, raw unchanged version and no evidence for real or outdoor navigation use.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its

evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 6: Label Quality. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use physical row cards and overlapping circles for flags. Supply blank card headings but no completed findings. Extension: explain how to record an evidenced seconds-to-minutes conversion while preserving the raw value, unit and version, and why a unit guess is not enough.

Require W4 counts 6,5,2,1,1 in the stated order, the Q overlap, acceptance of zero under this schema, and a card with source, unit, raw version and excluded use. Fresh retry: rows (L,3), (M,None), (N,-2), (N,-2), (O,0). Counts are 5 raw, 4 IDs, 1 missing, 2 negative and 1 extra ID occurrence. The negative and repeated-ID flags overlap on N. Do not infer whether the two N rows describe one event without checking the source.

Week 5 examines label definitions and disagreement. Carry forward the dataset card, source questions and explicit distinction between unknown and observed values. Teacher reference: <https://arxiv.org/>

Readiness and reassessment for Units conversions and data provenance

Use 60-second strips. Extension: compare results before and after rounding and explain why early rounding can change totals.

Independent W4 evidence: 4.0, 1.5 and 1.0 minutes; total 6.5 minutes. Retain original units and values. Record whether the learner needed no help, a hint or explanation. Fresh retry: 90 seconds, 2 minutes and 30 seconds become 1.5, 2, 0.5; total 4 minutes. Keep the three source records.

Plan the delayed return

Suggested return points in this level: ai-4/6, ai-4/8, ai-4/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Prepare, teach and assess

Teaching target: Inspect row meaning missing values and repeated records

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Explain each number in 5 4 2 1. How many extra ID

occurrences are there, and how many rows belong to a repeated-ID group?

Reasoned practice answer: The numbers mean 5 raw rows, 4 distinct IDs, 2 missing durations and 1 negative duration. There is 1 extra ID occurrence; 2 rows belong to the repeated-ID group B. These are different counts.

Check every learner before discussion: Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The card records evidence and gaps; its claims still need checking. Documentation does not prove suitability.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The teacher confirms B’s second row is an accidental duplicate. Require a prediction and a reason before comparing. Expected result: Remove that duplicate with a recorded reason; C still needs correction or explicit exclusion. Reason: Resolving one defect does not resolve a separate impossible measurement.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.